

CONVOLUTIONAL AUTOENCODER FOR ANOMALY DETECTION IN IMAGES

Manar Almudayfir
Raghad Alsnaie

Motivation and objectives

Motivation

Anomaly detection focuses on identifying data samples that deviate from normal patterns. In many real-world applications, such as industrial inspection, medical imaging, and cybersecurity, anomalous samples are rare or not available during training. This makes supervised learning approaches ineffective and motivates the use of unsupervised methods. Autoencoders are well-suited for this task because they learn to reconstruct normal data without requiring labeled anomalies. When trained only on normal samples, an autoencoder can accurately reconstruct normal patterns but produces higher reconstruction errors for anomalous inputs. This reconstruction error can therefore be used as a reliable anomaly score.

Motivation and objectives

objectives

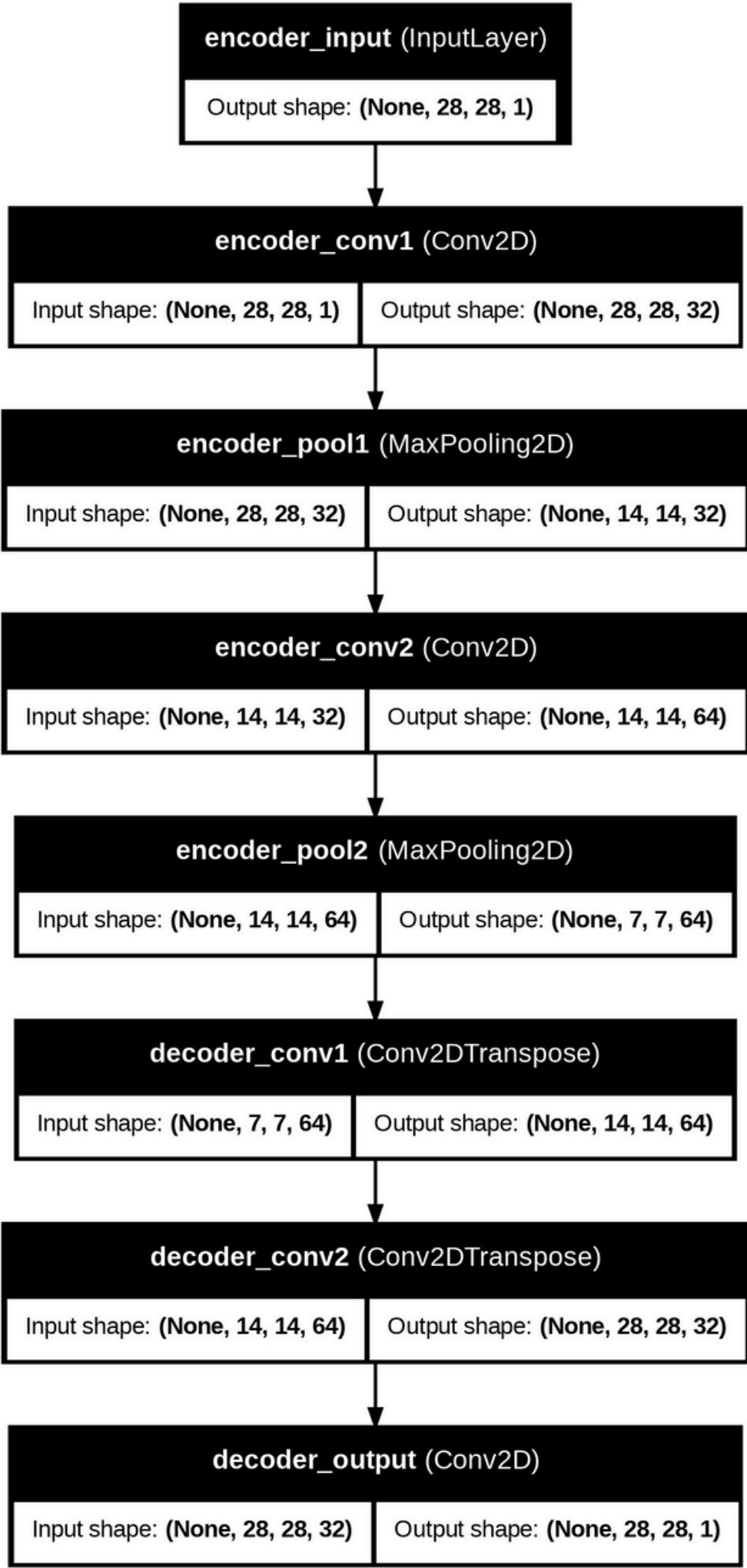
The objective of this project is to investigate the effectiveness of convolutional autoencoders for image based anomaly detection. Specifically, the project aims to design and compare multiple autoencoder architectures, including a baseline CAE, an improved CAE with skip connections, an attention-enhanced CAE, and a variational autoencoder. The goal is to analyze how architectural enhancements and probabilistic modeling impact reconstruction quality and anomaly detection performance.

Dataset summary

- The experiments were conducted using the Fashion-MNIST dataset, which consists of 28×28 grayscale images across ten clothing categories. For anomaly detection, the dataset was configured as a one-class problem, where T-shirt/top images were treated as normal samples and all other classes as anomalies. Only normal images were used during training.
- The images were normalized to the range [0, 1] and reshaped for convolutional processing. Multiple convolutional autoencoder architectures were implemented, including a baseline model, an improved model with batch normalization and skip connections, an attention-enhanced model using self-attention, and a variational autoencoder with a probabilistic latent space and KL divergence regularization.

architecture visualization

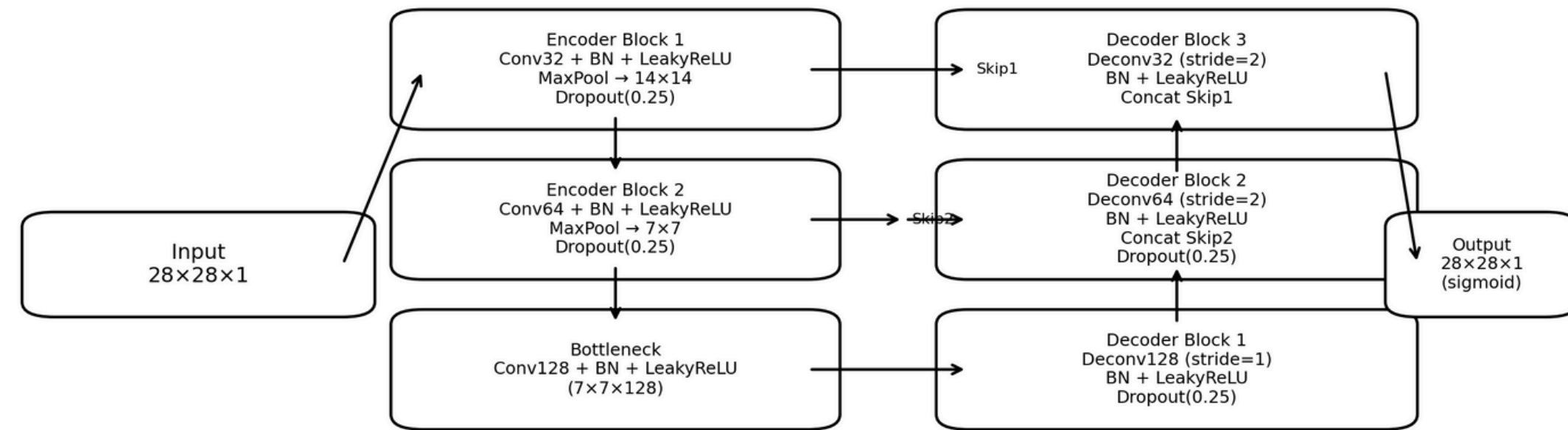
Baseline



architecture visualization

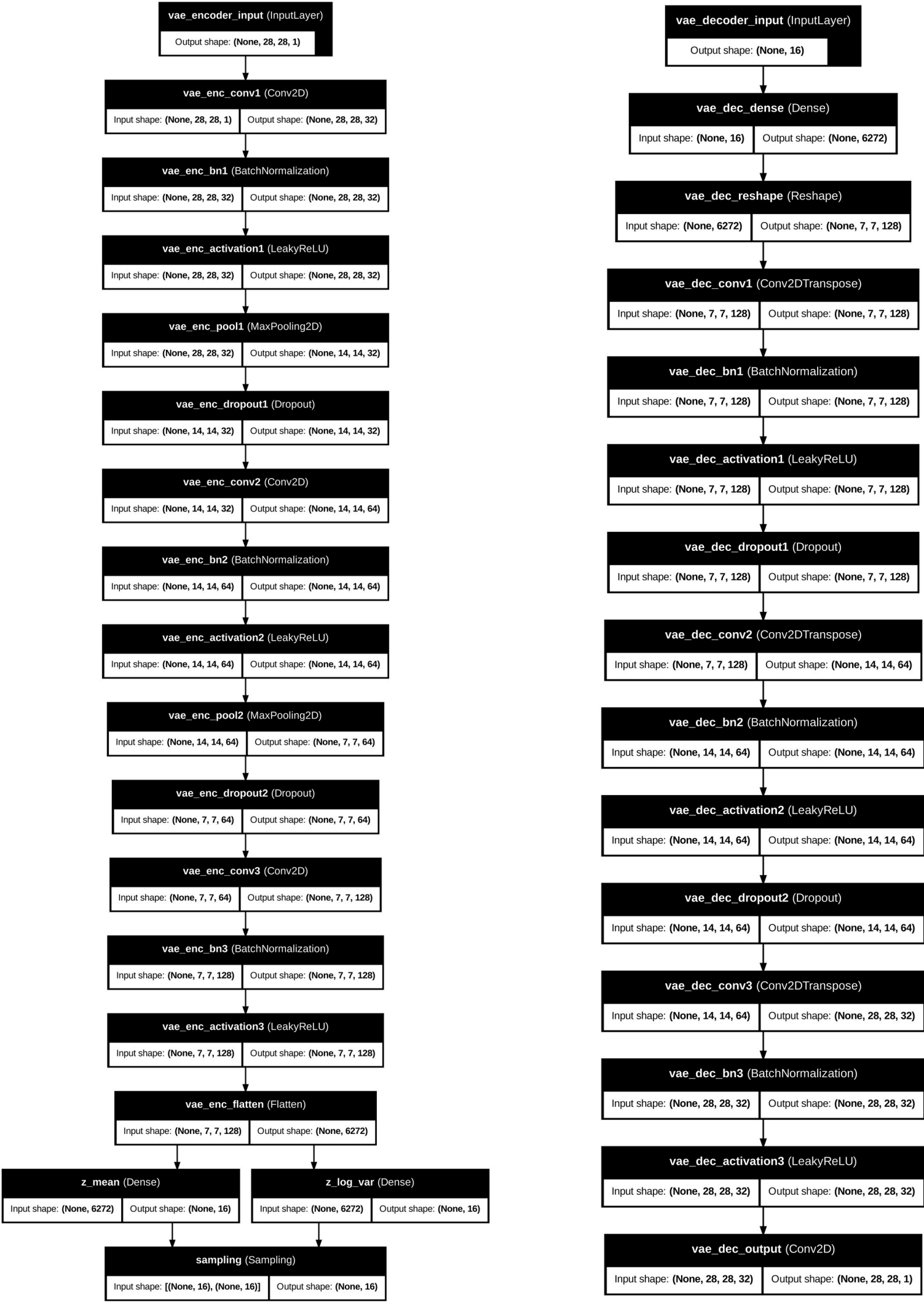
Improved

Improved_CAE (Block Diagram)



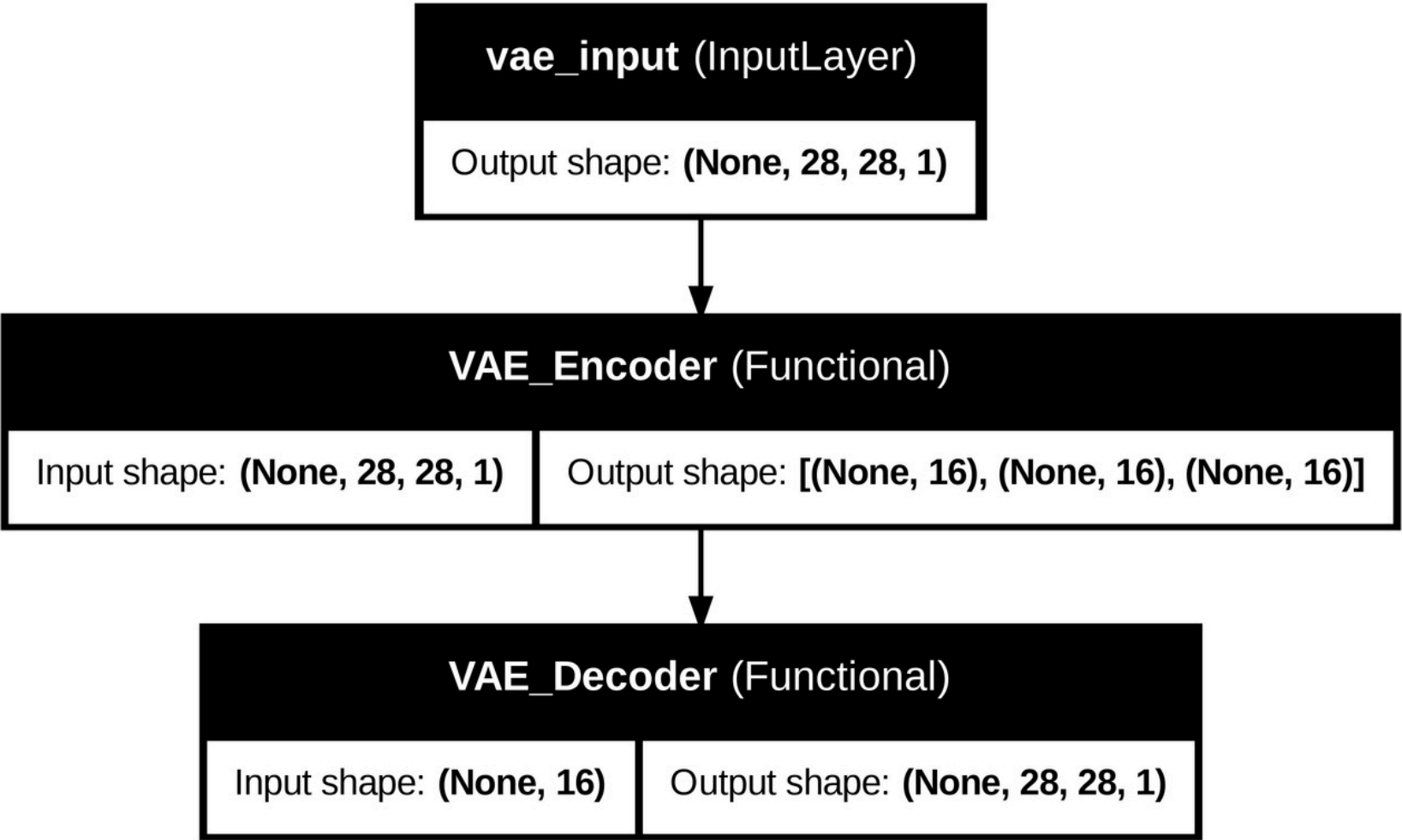
architecture visualization

VAE



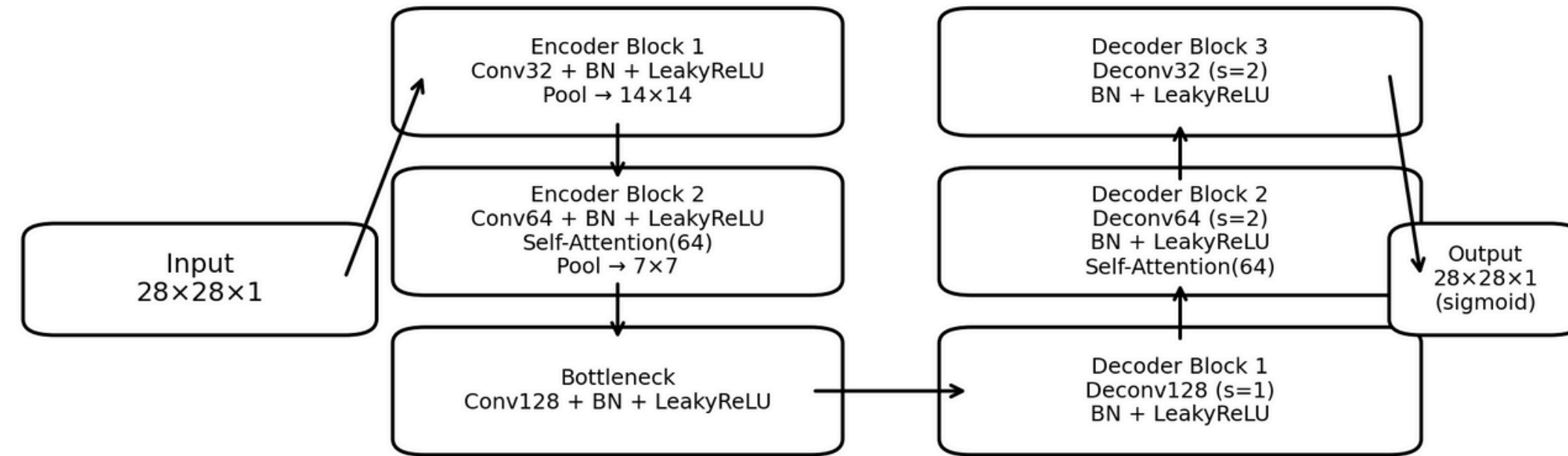
architecture visualization

VAE

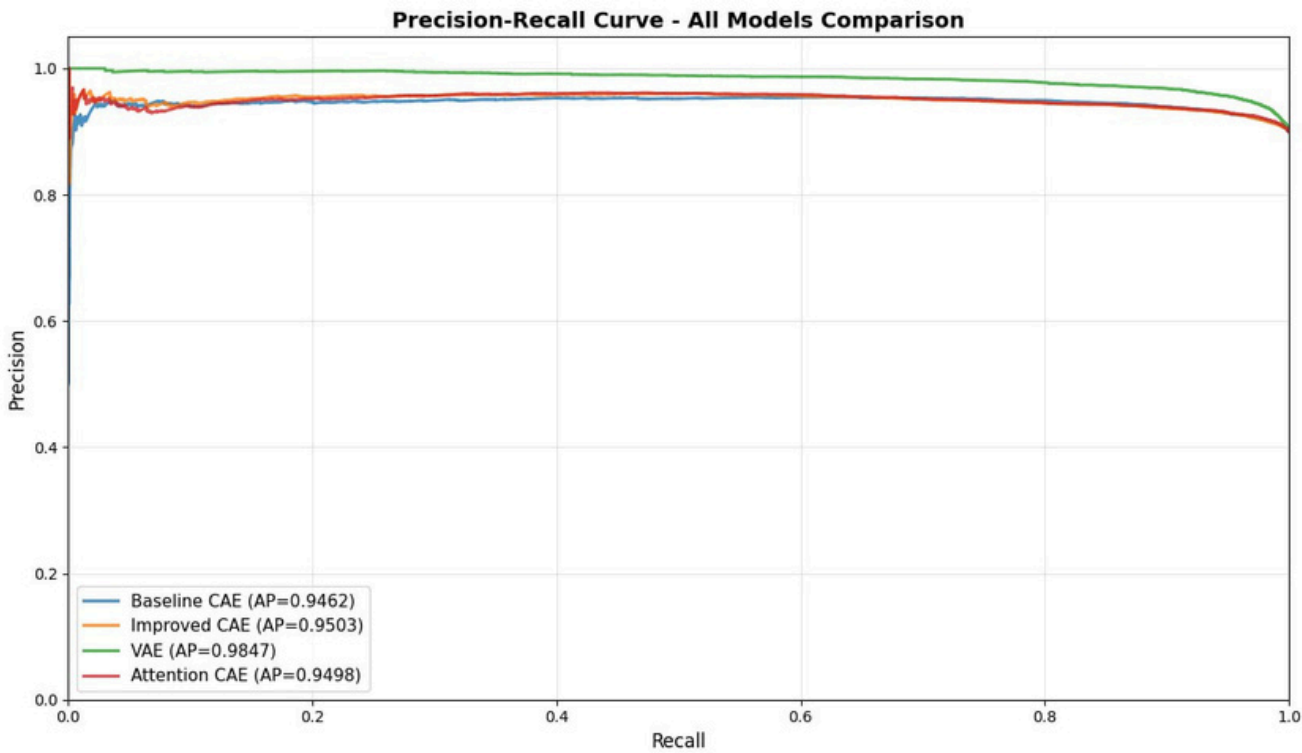
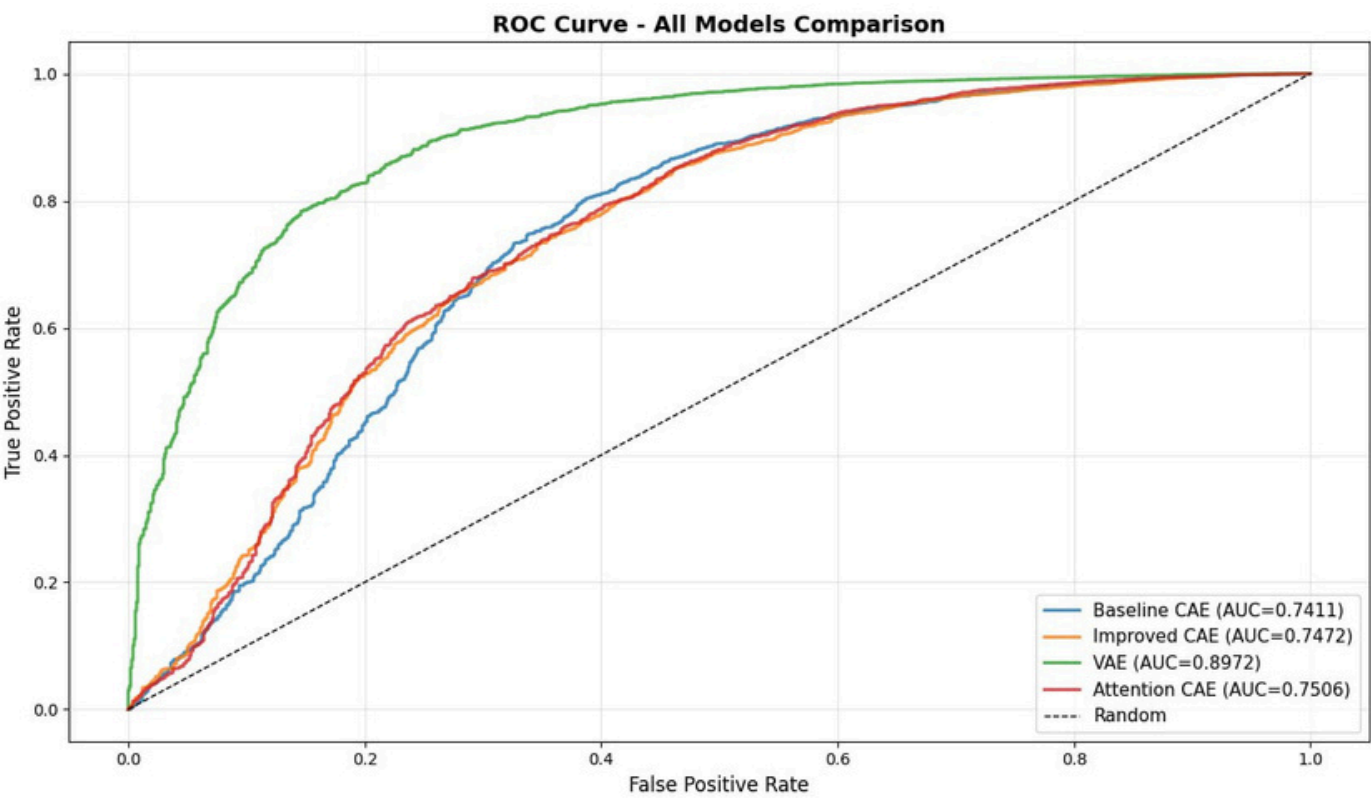


architecture visualization

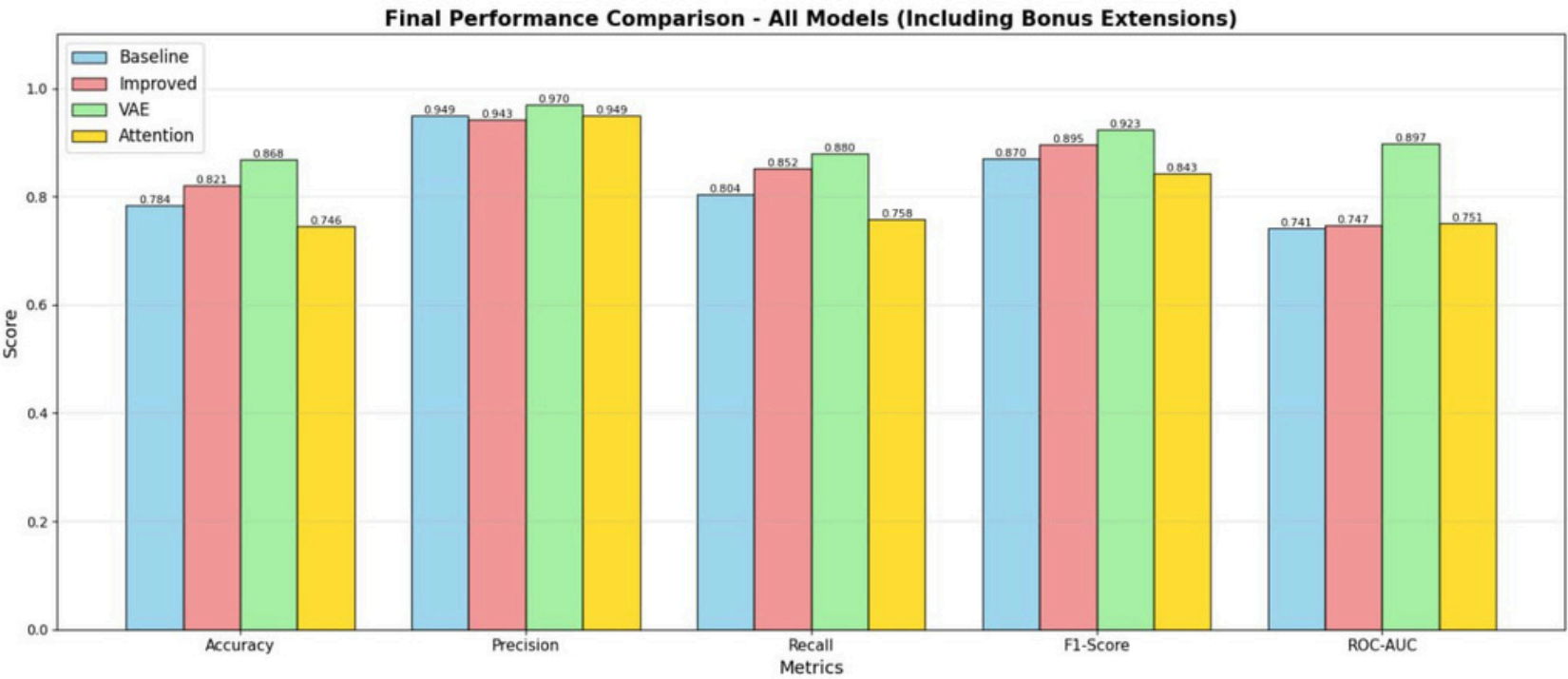
Attention



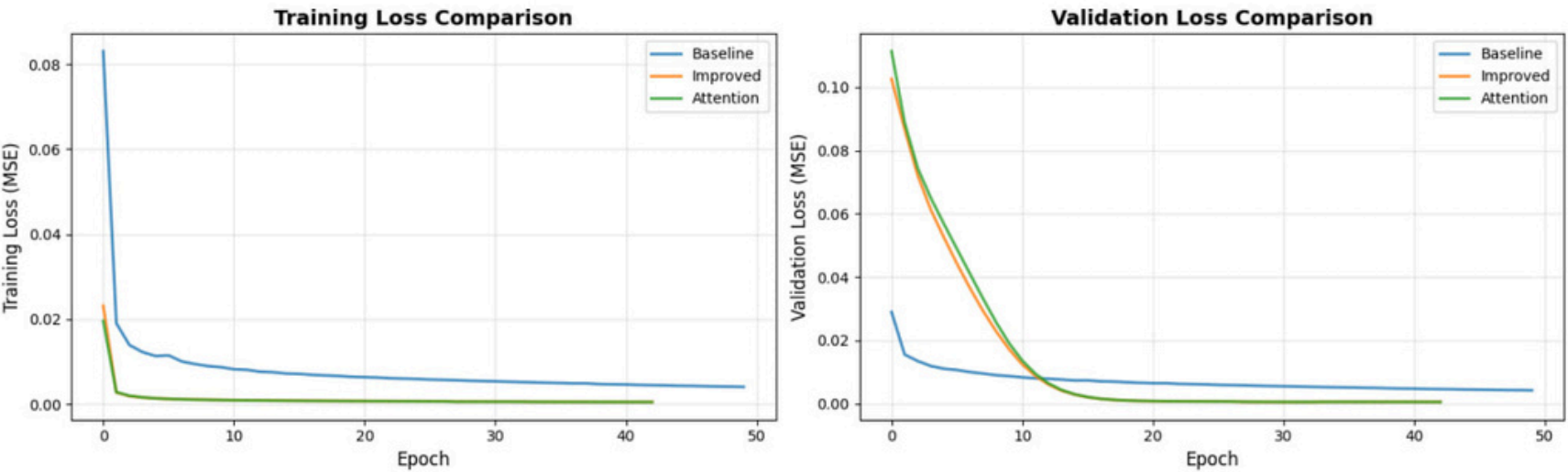
Results



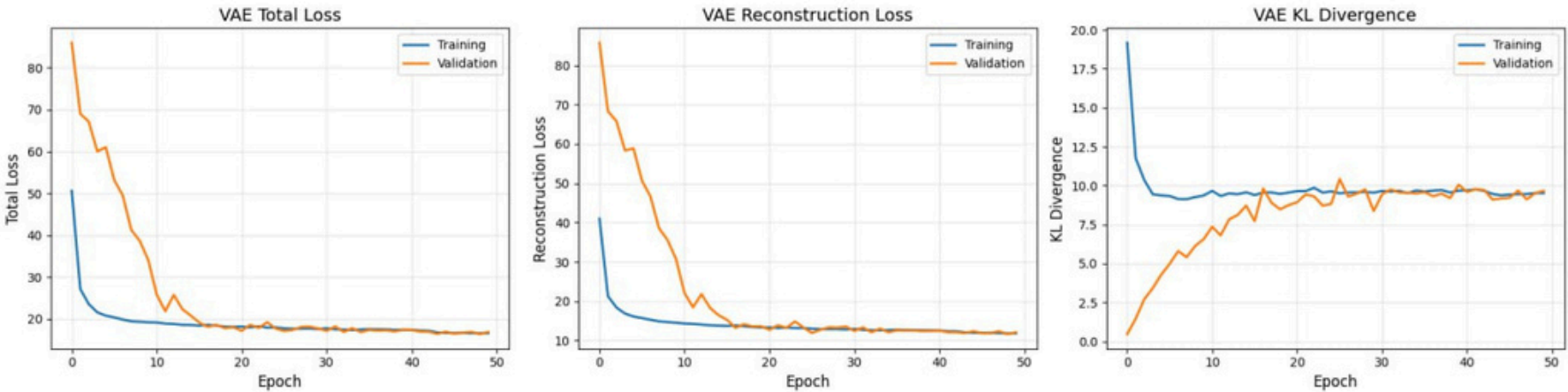
Results



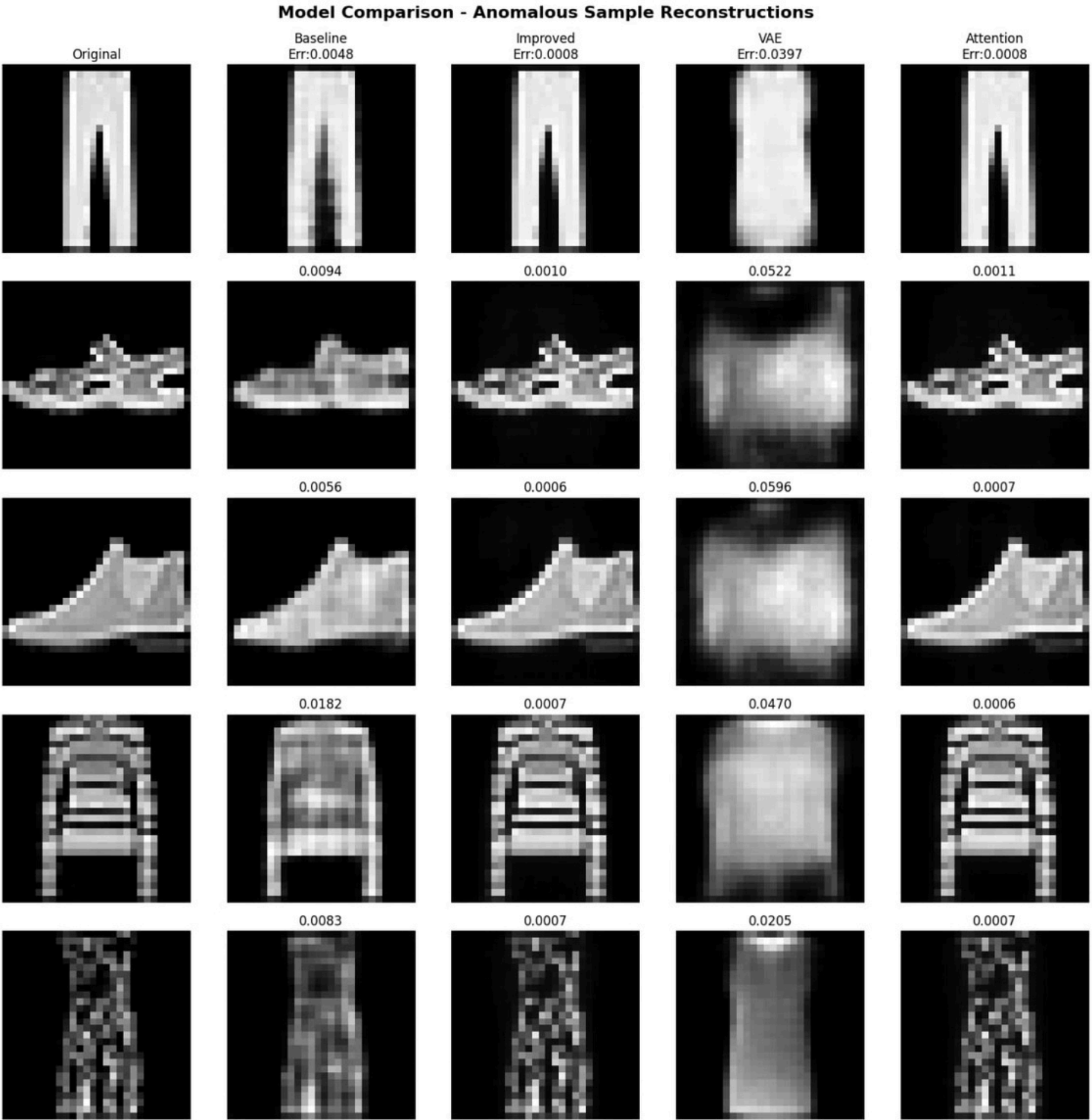
Model Training Comparison (CAE Models)



VAE Training History



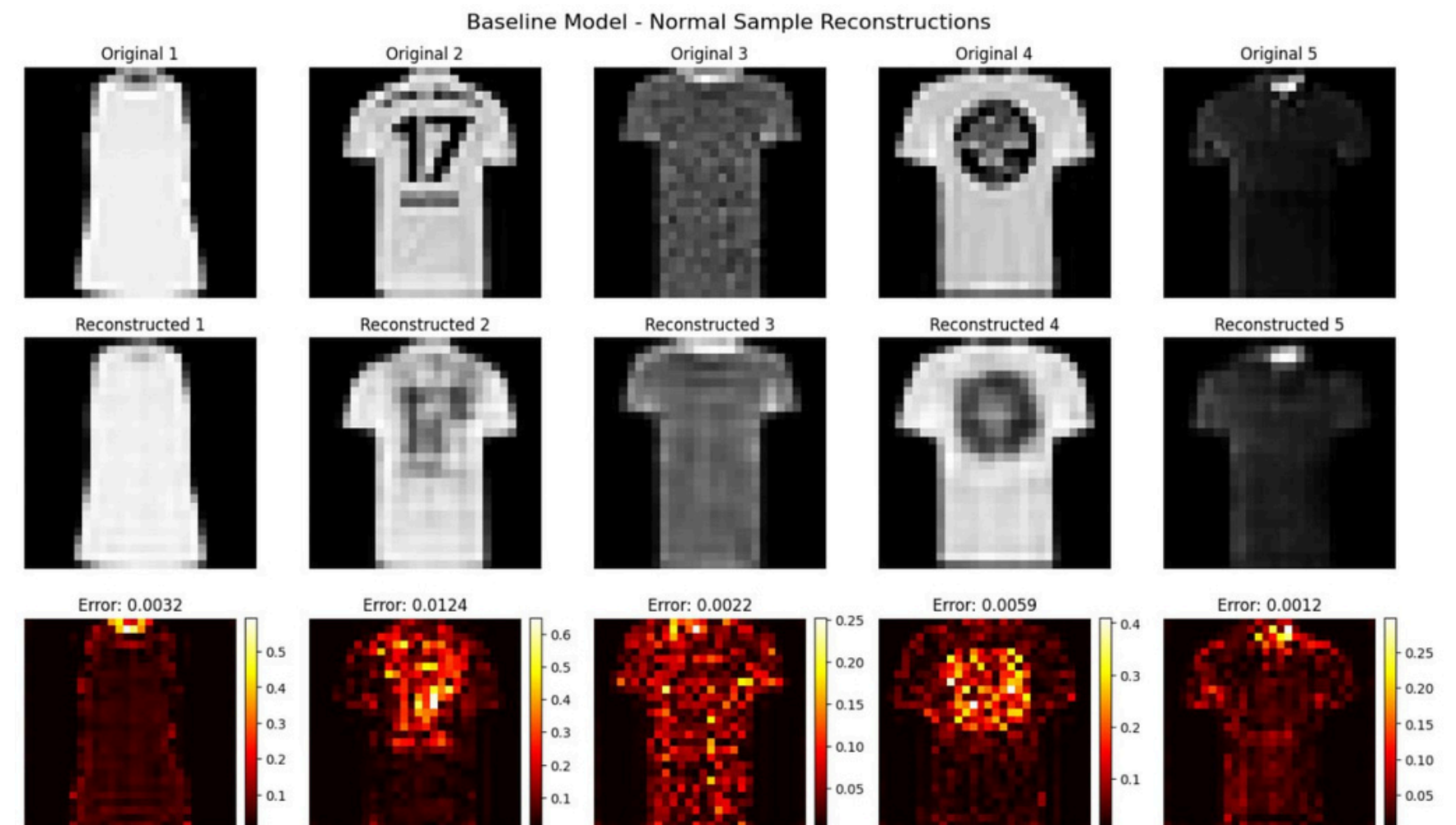
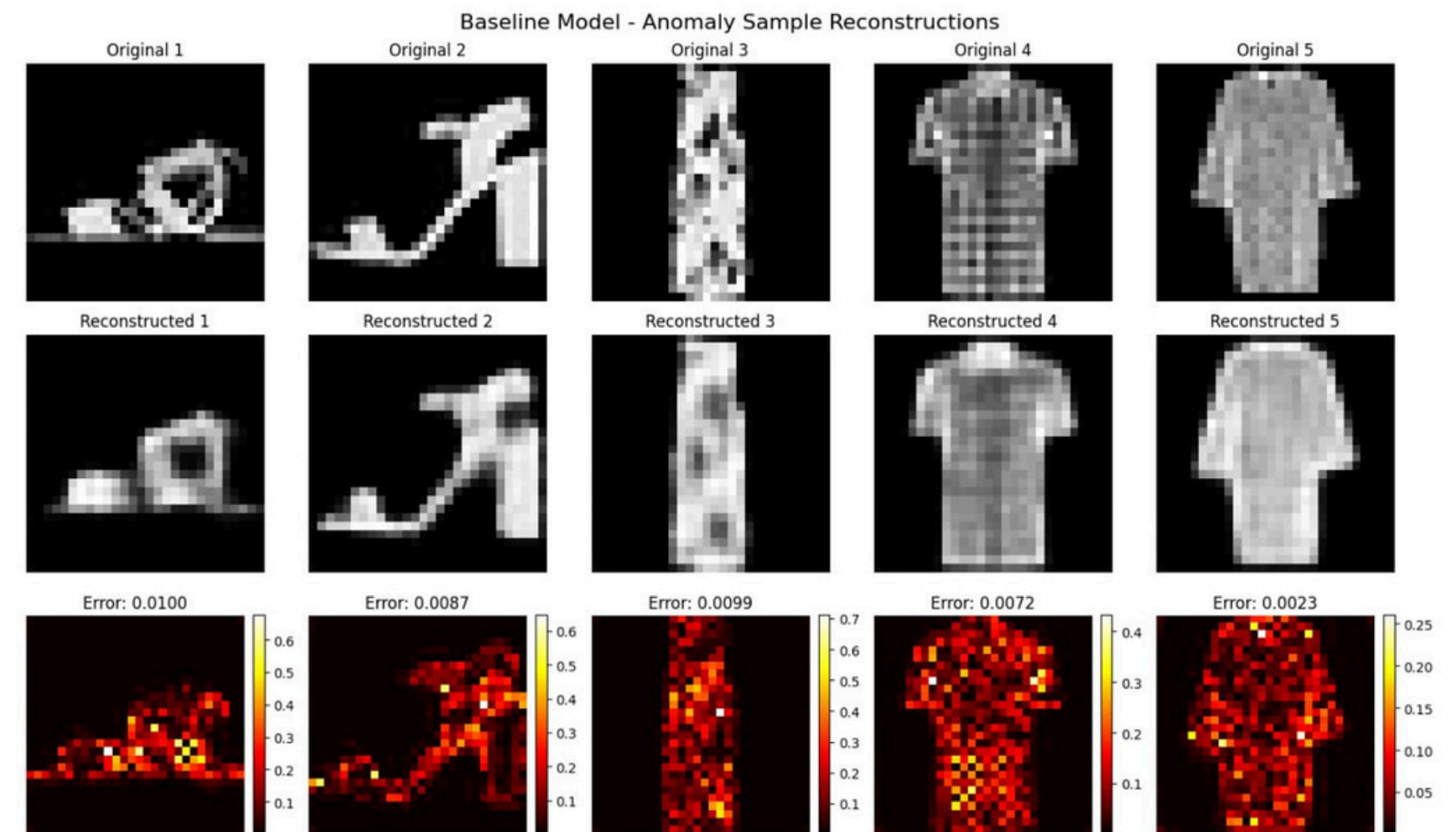
Anomaly detection examples



Normal detection examples



Anomaly Normal detection examples for Baseline



Challenges

- One of the main challenges observed in this project is the structural similarity between certain clothing categories. Anomalies that closely resemble T-shirts, such as pullovers and shirts, often resulted in lower reconstruction errors, making them harder to detect. This highlights a limitation of reconstruction-based anomaly detection methods when dealing with visually similar classes.
- Another challenge is the trade-off between reconstruction quality and detection sensitivity. Architectures that are highly expressive, such as attention-based models, may unintentionally learn to reconstruct anomalous patterns, reducing their effectiveness as anomaly detectors.

future improvements

- Future improvements could include combining attention mechanisms with variational autoencoders to benefit from both global feature modeling and probabilistic latent space constraints. Additionally, adversarial approaches such as GAN-based anomaly detection and transformer-based architectures could be explored to further enhance detection performance. Applying these models to higher-resolution and real-world industrial datasets would also provide more robust evaluation.

Thank You!

Any Questions?